TCubed Software Requirements Specification

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1. Introduction

1.1 Purpose

This section delineates the purpose of the software to be specified. This document specifies the software requirements for a web-based task tracking application. The system provides a visual, drag-and-drop interface for managing tasks, known as "Containers", across customizable workflow statuses.

1.2 Scope

This section describes the scope of the software under consideration. The system includes user authentication, workspace creation (Projects), workflow grouping (Boards), status lists (Buckets), and task management (Containers).

1.3 Product overview

1.3.1 Product perspective

This section defines the system's relationship to other related products. The application operates as a standalone web application accessible via modern web browsers (Chrome, Firefox, Safari, Edge).

1.3.2 Product functions

This section provides a summary of the major functions that the software will perform.

- User account creation and secure workspace setup.
- Creation and organization of project boards and status buckets.
- CRUD (Create, Read, Update, Delete) operations for task containers.

- Interactive drag-and-drop repositioning of containers across statuses.
- Archiving and trash management for discarded tasks.

1.3.3 User characteristics

This section describes those general characteristics of the intended groups of users of the product.

- Unregistered User: An individual who interacts with the system to create an ACCOUNT.
- Project Member: An authenticated user linked to a specific PROJECT via the PROJECT_MEMBER table. They have full rights to manage boards, buckets, and containers within that project.

1.3.4 Limitations

This section provides a general description of any other items that will limit the supplier's options. (To Be Determined)

1.4 Definitions

This section provides definitions for any words or phrases that have special meaning beyond normal dictionary definitions.

- Container: A task item that can be created, edited, and moved across different statuses.
- Bucket: A workflow column that holds containers and maintains a specific order.
- Board: A collection of buckets that organizes a specific initiative within a project.

2. References

This section provides a complete list of all documents referenced.

- ISO/IEC/IEEE 29148:2018 Systems and software engineering — Life cycle processes — Requirements engineering.

3. Requirements

3.1 Functions

This section defines the fundamental actions that have to take place in the software.

UC-1: Account Creation and Workspace Setup

- Preconditions: The user is on the application landing page.
- Main Flow:
    1. The user submits their FirstName, LastName, Email, and Password, to create an ACCOUNT.
    2. The system assigns a unique UserId.
    3. The user creates a new PROJECT, providing a Name and Description.
    4. The system links the UserId to the new ProjectId inside the PROJECT_MEMBER table.
- Postconditions: A new user account and project are stored in the database, and the user is routed to the project dashboard.

UC-2: Defining Boards and Status Buckets

- Preconditions: The user is authenticated and viewing an active PROJECT.
- Main Flow:
    1. The user creates a BOARD to organize a specific initiative.
    2. The system associates the BOARD with the current ProjectId.

3. The user creates multiple workflow columns, defined as a BUCKET in the system (e.g., "To Do", "In Progress").
4. The system links each BUCKET to the BoardId.
5. The system assigns a sequential Position integer to each BUCKET to maintain visual order.

- Postconditions: A new board containing ordered status buckets is rendered on the screen.

UC-3: Container Management (Creation & Editing)

- Preconditions: The user is viewing a board that contains at least one BUCKET.
- Main Flow:
    1. The user clicks to add a new Container inside a specific bucket.
    2. The user inputs a Title and Body for the container.
    3. The system inserts a new record into the TASK table.
    4. The system links the new task to the appropriate BucketId.
    5. The system generates a CreatedAt timestamp and assigns a Position to place it at the bottom of the list.
- Postconditions: The container appears in the selected bucket, and the database stores its state.

UC-4: Drag-and-Drop Repositioning

- Preconditions: The board contains multiple buckets and containers.
- Main Flow:
    1. The user clicks and drags a Container from its current bucket into a new bucket.
    2. The user drops the Container in a specific visual order between other existing containers.

3. The frontend UI immediately displays the Container in its new location.
4. The system asynchronously sends a request to update the database.
5. The system updates the Container's BucketId in the TASK table to match the new bucket.
6. The system updates the Position values of all affected records in the TASK table to reflect the new sorting order.
7. The system updates the UpdatedAt timestamp for the modified Container.

- Exceptions: If the database update fails, the frontend UI reverts the Container to its original position and alerts the user.

UC-5: Moving Containers to Trash/Archive

- Preconditions: A "Trash" or "Archive" bucket has been created on the board.
- Main Flow:
    1. The user drags a Container into the designated Trash BUCKET.
    2. The system updates the BucketId of the TASK to match the Trash bucket.
- Postconditions: The Container is removed from the active workflow but remains in the database until a user explicitly issues a command to permanently delete the contents of the Trash bucket.

3.2 Performance requirements

This section specifies both the static and the dynamic numerical requirements placed on the software. UI updates resulting from drag-and-drop actions must render in under 100 milliseconds locally to ensure a fluid experience, while SQL database updates to Position and BucketId execute in the background.

3.3 Usability requirements

This section defines usability and quality in use requirements and objectives. (To Be Determined)

3.4 Interface requirements

This section defines all inputs into and outputs from the software system. (To Be Determined)

3.5 Logical database requirements

This section specifies the logical requirements for any information that is to be placed into a database. The application operates on a client-server model using a relational SQL database backend defined by a specific hierarchy: ACCOUNT, PROJECT, PROJECT_MEMBER, BOARD, BUCKET, and TASK. The backend must enforce foreign key constraints ensuring that if a BOARD is deleted, all associated BUCKET records are dropped. Consequently, if a BUCKET is deleted, all associated TASK records must be handled (either cascaded, deleted, or moved to a fallback bucket).

3.6 Design constraints

This section specifies constraints on the system design imposed by external standards, regulatory requirements or project limitations. (To Be Determined)

3.7 Software system attributes

This section specifies the required attributes of the software product.

- Security: Server-side logic must verify that the requester is a valid PROJECT_MEMBER before executing any UPDATE or INSERT commands on a TASK, BUCKET, or BOARD within that project.

3.8 Supporting information

(To Be Determined)

4. Verification

This section provides the verification approaches and methods planned to qualify the software.

(To Be Determined)

5. Appendices

5.1 Assumptions and dependencies

This section lists each of the factors that affect the requirements stated in the SRS. (To Be Determined)

5.2 Acronyms and abbreviations

This section spells out or defines all acronyms and abbreviations used in the documents.

- SRS: Software Requirements Specification
- UI: User Interface
- SQL: Structured Query Language
- CRUD: Create, Read, Update, Delete